

College: Guru Nanak First Grade College Bidar
Department: BCA
Semester: IV
Course Code: SEPBCAP10
Course Title: Java Programming Lab

PART-A:

1. Java Program to Perform Arithmetic Operations

```
import java.util.Scanner;

public class Arithmetic {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter First Number: ");

        int a = sc.nextInt();

        System.out.print("Enter Second Number: ");

        int b = sc.nextInt();

        System.out.println();

        System.out.println("Arithmetic Operations: ");

        System.out.println("-----");

        System.out.println("Addition = " + (a + b));

        System.out.println("Subtraction = " + (a - b));

        System.out.println("Multiplication = " + (a * b));

        System.out.println("Division = " + (a / b));

        sc.close();

    }

}
```

Output:

Enter First Number: 2
Enter Second Number: 2
Arithmetic Operations:

Addition = 4
Subtraction = 0
Multiplication = 4
Division = 1

2. Java program to Read n Subject Marks and Display Maximum and Average

```
import java.util.Scanner;

public class SubjectMarks {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of subjects: ");

        int n = sc.nextInt();

        int sum = 0, max = 0, subj = 1;

        for(int i = 1; i <= n; i++) {

            System.out.print("Enter Marks-" + subj + ": ");

            int mark = sc.nextInt();

            sum += mark;

            subj++;

            if(mark > max)

                max = mark;

        }

        System.out.println("Maximum = " + max);

        System.out.println("Average = " + (sum/(double)n));
```

```
sc.close();  
  
}  
  
}
```

Output:

```
Enter number of subjects: 5  
Enter Marks-1: 20  
Enter Marks-2: 30  
Enter Marks-3: 40  
Enter Marks-4: 50  
Enter Marks-5: 60  
Maximum = 60  
Average = 40.0
```

3. Java program to find largest of three numbers.

```
import java.util.Scanner;  
  
public class LargestOfThree {  
  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter First Number: ");  
  
        int a = sc.nextInt();  
  
        System.out.print("Enter Second Number: ");  
  
        int b = sc.nextInt();  
  
        System.out.print("Enter Third Number: ");  
  
        int c = sc.nextInt();  
  
        int largest = a;  
  
        if(b > largest)  
            largest = b;  
  
        if(c > largest)
```

```
        largest = c;
System.out.println();
System.out.println("Largest number is: " + largest);
sc.close();
    }
}
```

Output:

```
Enter First Number: 20
Enter Second Number: 30
Enter Third Number: 50

Largest number is: 50
```

4. Java program to print first n terms of Fibonacci sequence.

```
import java.util.Scanner;

public class FibonacciSequence {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter the number of terms: ");

        int n = scanner.nextInt();

        int firstTerm = 0, secondTerm = 1;

        System.out.println("Fibonacci Sequence up to " + n + " terms:");

        for (int i = 1; i <= n; ++i) {

            System.out.print(firstTerm + (i < n ? ", " : ""));

            // Compute the next term

            int nextTerm = firstTerm + secondTerm;
```

```

        firstTerm = secondTerm;

        secondTerm = nextTerm;
    }

    scanner.close();
}
}

```

Output:

Enter the number of terms: 8

Fibonacci Sequence up to 8 terms:

0, 1, 1, 2, 3, 5, 8, 13

5.Java program to Swap Two Numbers Without Third Variable.

```

import java.util.Scanner;

public class SwapTwoNumbers {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter two numbers: ");

        System.out.println("-----");

        System.out.print("Enter First Number: ");

        int firstNum = sc.nextInt();

        System.out.print("Enter Second Number: ");

        int secondNum = sc.nextInt();

        firstNum = firstNum + secondNum;

        secondNum = firstNum - secondNum;
    }
}

```

```

        firstNum = firstNum - secondNum;

        System.out.println();

        System.out.println("Numbers After swapping:");

        System.out.println("-----");

        System.out.println("First Number = " + firstNum);

        System.out.println("Second Number = " + secondNum);

    }

}

```

Output:

```

Enter two numbers:
-----
Enter First Number: 20
Enter Second Number: 50

Numbers After swapping:
-----
First Number = 50
Second Number = 20

```

6. Java program to find sum of first n even numbers divisible by 3.

```

import java.util.Scanner;

public class SumEvenDivisibleBy3 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter value of n: ");
        int n = sc.nextInt();

        int sum = 0;
        int count = 0;
        int num = 6;
    }
}

```

```
while (count < n) {  
    sum += num;  
    num += 6;  
    count++;  
}
```

```
System.out.println("Sum = " + sum);
```

```
sc.close();  
}  
}
```

Output:

Enter value of n: 3
Sum = 36

7. Java program to check given number is Armstrong or not.

```
import java.util.Scanner;
```

```
public class Armstrong {
```

```
    static boolean isArmstrong(int num) {
```

```
        int temp = num;
```

```
        int digits = 0;
```

```
        int sum = 0;
```

```
        int countTemp = num;
```

```
        while (countTemp > 0) {
```

```
            countTemp /= 10;
```

```
            digits++;
```

```
        }
```

```
        temp = num;
```

```

while (temp > 0) {
    int lastDigit = temp % 10;

    int powerValue = 1;

    for (int i = 0; i < digits; i++) {
        powerValue *= lastDigit;
    }

    sum += powerValue;

    temp /= 10;
}

return sum == num;
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter a number: ");

    int number = sc.nextInt();

    if (isArmstrong(number)) {
        System.out.println(number + " is an Armstrong number.");
    } else {
        System.out.println(number + " is not an Armstrong number.");
    }
}
}

```

Output:

```

Enter a number: 2
2 is an Armstrong number.

```


8. Java program to read and display student information using class.

```
import java.util.Scanner;

class StudentInformation {

    int rollNo;

    String name;

    int marks;

    void readData() {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter Student Information: ");

        System.out.println("-----");

        System.out.print("Enter Roll Number: ");

        rollNo = sc.nextInt();

        sc.nextLine();

        System.out.print("Enter Name: ");

        name = sc.nextLine();

        System.out.print("Enter Marks: ");

        marks = sc.nextInt();

    }

    void displayData() {

        System.out.println("\nStudent Information: ");

        System.out.println("-----");

        System.out.println("Roll Number: " + rollNo);

        System.out.println("Name: " + name);

        System.out.println("Marks: " + marks);

    }

}
```

```

    }
}

class StudentInfoTest {

    public static void main(String[] args) {

        StudentInformation si = new StudentInformation();

        si.readData();

        si.displayData();

    }

}

```

Output:

Enter Student Information:

Enter Roll Number: 10012

Enter Name: Ram

Enter Marks: 450

Student Information:

Roll Number: 10012

Name: Ram

Marks: 450

9. Java program on constructor.

```

class Database {

    private String database;

    private String userName;

    private String password;

```

```

Database() {
    database = "ORACLE";
    userName = "system";
    password = "manager";
}

void displayDatabase() {
    System.out.println("Database Information");
    System.out.println("-----");
    System.out.println("Database: " + database);
    System.out.println("User-Name: " + userName);
    System.out.println("Password: " + password);
}

public static void main(String[] args) {
    Database db = new Database();
    db.displayDatabase();
}
}

```

Output:

```

Database Information
-----
Database: ORACLE
User-Name: system
Password: manager

```

10. Java program on method overloading.

```

class PaymentGateway {
    void makePayment(long cardNumber, int cvv) {

```

```

        System.out.println("Processing payment for Card: " + cardNumber);
    }

    void makePayment(String upid) {

        System.out.println("Processing payment via UPI ID: " + upid);
    }

    void makePayment(long cardNumber, String promoCode) {

        System.out.println("Processing payment for Card with promo code: " + promoCode);
    }
}

class PaymentGatewayTest {

    public static void main(String[] args) {

        PaymentGateway payment = new PaymentGateway();

        payment.makePayment(1234567812345678L, 489);

        payment.makePayment("ram@upi");

        payment.makePayment(1234567812345678L, "SAVE20");
    }
}

```

Output:

```

Processing payment for Card: 1234567812345678
Processing payment via UPI ID: ram@upi
Processing payment for Card with promo code: SAVE20

```